

BioMesh Algorithms and Data Structures

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1 Algorithms

1.1 Fiber Grid Generation

For generating 1D fibers we need to answer 3 fundamental questions:

- Where does a fiber begin ?
- Where does a fiber end ?
- What are the bounds of the muscle ?

The fiber generation workflow follows these steps sequentially:

- Cell Classification
- Seed Plane Generation
- Seed Point Generation
- Fiber generation
- Fiber grid generation

The **Seed Plane Generation** routine provides the surface on which the **Seed Points** are computed. This surface contains the Seed Points, these points mark the beginning of a fiber.

The **Cell Classification** routine is the solution to the final 2 questions. The classification contains data about the muscle bounds. The '1' cells mark the end of a fiber. Refer to the **Algorithms** section for detailed explanation.

1.1.1 Description

- **Cell Classification**

The first step is to identify the bounds of vector field. It represents the region within the structured grid where the fibers are allowed to flow freely.

The VTK cells are classified as '0', '1', or '2' cell. The definitions are as follows:

- '0' cell : The VTK cells which do not have vectors on any corners.
- '1' cell : The VTK cells which have vectors on some but not all corners.

- '2' cell : The VTK cells which have vectors on all corners.

The '1' cells represent the bounds of the vector field. The fibers are allowed to flow freely in the '1' and '2' cells.

Source files: `biomesh_cell_table.hpp`, `biomesh_cell_table.cpp`

Algorithm:

Require: Structured grid G

Ensure: Classification label $cell_type$ for each cell $c \in G$

```

1: for all cells  $c \in G$  do
2:    $V \leftarrow \emptyset$  ▷ Initialize vector set for cell corners
3:   for all corners  $v$  of  $c$  do
4:     Obtain vector  $\mathbf{v}$  at corner  $v$ 
5:      $V \leftarrow V \cup \{\mathbf{v}\}$  ▷ Add vector to set
6:   end for
7:   if  $\|\mathbf{v}\| = 0 \ \forall \mathbf{v} \in V$  then
8:      $cell\_type \leftarrow 0$  ▷ All zero vectors
9:   else if  $\|\mathbf{v}\| \neq 0 \ \forall \mathbf{v} \in V$  then
10:     $cell\_type \leftarrow 2$  ▷ All non-zero vectors
11:   else
12:     $cell\_type \leftarrow 1$  ▷ Mixed zero and non-zero vectors
13:   end if
14: end for

```

- **Seed Plane Generation**

The second step is identify the beginning of the fibers. The **Seed Plane** splits the vector field in two parts. Naturally, the **Seed Plane** intersects the structured grid cells. The plane-cell intersection is a surface triangulation. The seed points will be computed on this surface.

Source files: `biomesh_seed_plane.hpp`, `biomesh_seed_plane.cpp`

Algorithm:

Require: Structured grid G , cell classification C

Ensure: Triangle set T containing valid intersections

```

1:  $T \leftarrow \emptyset$  ▷ Initialize triangle set
2: Construct plane  $P$ 
3:  $S \leftarrow P \cap G$  ▷ Surface triangulation S from intersecting P with G
4: for all triangles  $t \in S$  do
5:    $c \leftarrow \text{centroid}(t)$ 
6:    $cell \leftarrow \text{cell containing } c$ 
7:   if  $c \notin G$  then
8:     Skip  $t$ 
9:   end if
10:  if  $C(cell) = 2$  then
11:     $T \leftarrow T \cup \{t\}$  ▷ Add triangle if cell type indicates valid region
12:  end if
13: end for

```

- **Seed Point Generation**

The third step is compute the seed points. These points are the initial conditions for the numeric integration scheme. The Random Sampling technique is used to generate the points.

First, the center of the triangles are computed. Then the triangles whose centers are within the bounds of the '2' cells are checked. If their centers are within the '2' cell bounds then the center is flagged as a seed point.

Source files: `biomesh_seeder.hpp`, `biomesh_seeder.cpp`

Algorithm:

Require: seed plane P , seed count C

Ensure: Seed vertex set S

```

1:  $S \leftarrow \emptyset$                                 ▷ Initialize set of seed vertices
2:  $A \leftarrow \emptyset$                                 ▷ Cumulative area array
3:  $TotalArea \leftarrow 0$ 
4: for all triangles  $t \in P$  do
5:    $TotalArea \leftarrow TotalArea + Area(t)$ 
6:    $A \leftarrow A \cup \{TotalArea\}$ 
7: end for
8: Initialize random number generator  $R$ 
9: for  $i = 0$  to  $C - 1$  do
10:   $r \leftarrow$  random value from  $R$  scaled by  $TotalArea$ 
11:  Find triangle  $t \in P$  corresponding to index  $r$  in  $A$ 
12:   $u \leftarrow$  random value from  $R$                                 ▷ Barycentric weights
13:   $v \leftarrow$  random value from  $R$ 
14:   $w \leftarrow 1 - u - v$ 
15:  Let  $p_0, p_1, p_2$  be vertices of triangle  $t$ 
16:   $seed \leftarrow w \cdot p_0 + u \cdot p_1 + v \cdot p_2$ 
17:   $S \leftarrow S \cup \{seed\}$                                 ▷ Add generated seed to set
18: end for

```

- **Fiber Generation**

The fourth step is to execute the Runge-Kutta4 scheme to trace the fibers.

Source files: `biomesh_fiber3d.hpp`, `biomesh_fiber3d.cpp`

Algorithm:

Require: Structured grid G , cell classification C , configuration parameters T , direction D

Ensure: Fiber f generated from seed vertex s

```

1:  $t_{start} \leftarrow 0$ 
2:  $\Delta t \leftarrow vertex\_width(T)$ 
3:  $vc \leftarrow vertex\_count(T)$ 
4:  $s \leftarrow$  seed vertex
5:  $t \leftarrow 0$ 
6:  $t_{max} \leftarrow time\_step\_max(T)$ 
7: Initialize Runge-Kutta 4 integrator  $I$ 
8: if  $strategy(T) = \text{static}$  then
9:    $adaptivity \leftarrow \text{false}$ 
10: else if  $strategy(T) = \text{adaptive}$  then

```

```

11:   adaptivity  $\leftarrow$  true
12: end if
13:  $N_{\text{adaptive}} = 0$ 
14:  $N_{\text{adaptive}}^{\text{max}} \leftarrow \text{adaptive\_steps\_max}(T)$ 
15:  $cid \leftarrow \text{Index}(\{c \in G \mid s \in c\})$ 
16: while ( $C(cid) = 1 \vee C(cid) = 2$ ) &  $t < t_{\text{max}}$  do
17:   if  $\neg \text{adaptivity} \vee (\text{adaptivity} \wedge N_{\text{adaptive}}^{\text{max}} \text{ reached})$  then
18:      $v_{\text{curr}} \leftarrow s$ 
19:     if  $v_{\text{curr}} \neq \text{last}(f)$  then
20:        $f \leftarrow f \cup \{v_{\text{curr}}\}$  ▷ Append vertex to fiber
21:     end if
22:   end if
23:    $cid \leftarrow \text{Index}(\{c \in G \mid v_{\text{curr}} \in c\})$ 
24:    $v_{\text{previous}} \leftarrow v_{\text{curr}}$ 
25:    $v_{\text{curr}} \leftarrow I(G, v)$  ▷ Advance one integration step
26:   if  $v_{\text{curr}} \neq v_{\text{previous}}$  then
27:      $v_{\text{diff}} \leftarrow v_{\text{new}} - v_{\text{previous}}$ 
28:     if  $D \leftarrow 1$  then
29:        $v_{\text{diff}} \leftarrow -v_{\text{diff}}$ 
30:     end if
31:   end if
32:   if  $C(i) = 0$  then
33:     if  $\text{adaptivity}$  then
34:       if  $N_{\text{adaptive}}^{\text{max}} = N_{\text{adaptive}}$  then
35:         break
36:       else
37:          $\Delta t \leftarrow \frac{\Delta t}{2}$  ▷ Reduce step size
38:          $N_{\text{adaptive}} \leftarrow N_{\text{adaptive}} + 1$ 
39:       end if
40:     else
41:       break
42:     end if
43:   end if
44:    $v_{\text{curr}} \leftarrow v_{\text{previous}} + \frac{v_{\text{diff}}}{\|v_{\text{diff}}\|}$ 
45:
46:    $t_{\text{start}} \leftarrow t_{\text{start}} + \Delta t$ 
47: end while

```

• Fiber Grid Generation

The final step is to execute the fiber generation routine for every **Seed Point** to generate the **Fiber Grid**. This is done in two steps (Forward and backward). For every seed point, the fiber is computed in the direction of the vector field flow (Forward) and then in the opposite direction of vector field flow (backward).

Source files: **biomesh_fiber_grid.hpp**, **biomesh_fiber_grid.tpp**

Algorithm:

Require: Structured grid G , seed set S

Ensure: Fiber grid F

- 1: $F \leftarrow \emptyset$ ▷ Initialize fiber grid
- 2: **for all** seeds $s \in S$ **do**

```

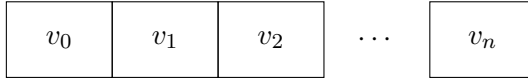
3:    $f \leftarrow \text{generate\_fiber}(s, G, \text{backward})$ 
4:    $f \leftarrow \text{reverse}(f)$  ▷ reverse the order for vertices
5:    $f \leftarrow \text{generate\_fiber}(s, G, \text{forward})$ 
6:    $F \leftarrow F \cup \{f\}$  ▷ Insert generated fiber into grid
7: end for

```

2 Data Structures

2.1 fiber:

A single fiber is represented as a contiguous array of vertices.



2.2 fiber_grid:

All fibers are stored using an 'array-of-arrays' data structure.

